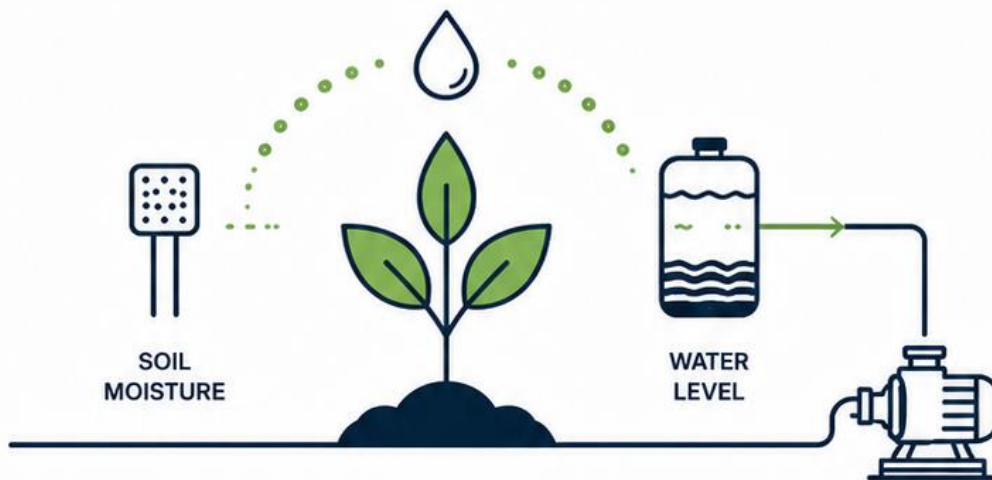


SMART IRRIGATION SYSTEM

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Embedded Real-Time Irrigation Control System



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Embedded Systems | RTOS | Smart Control Systems



PROJECT FOCUS

Real-Time Irrigation Control • Sensor Monitoring
Safety Logic • Embedded System Design



PLATFORM

STM32F103C8



DOCUMENT VERSION

Version 1.0



DATE

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Designed as a self-directed independent engineering portfolio project focused on real-time control, reliability, safety, and professional engineering documentation.

1. Document Control

Item	Description
Document Title	Smart Irrigation System – Software Requirements Specification
Document Type	SRS
Version	1.0
Project Platform	STM32F103C8
Software Environment	STM32CubeIDE, STM32CubeMX, CMSIS-RTOS V2
Purpose	Define the detailed software requirements required to implement and test the Smart Irrigation System.

2. Introduction

This Software Requirements Specification defines the software behavior required for the Smart Irrigation System.

The document translates the customer-level requirements defined in the CRS into detailed software requirements that can be implemented, verified, and validated.

The software shall monitor soil moisture, detect tank availability, control the pump safely, support manual watering, detect faults, provide local and serial status reporting, and use a real-time operating system for concurrent task execution.

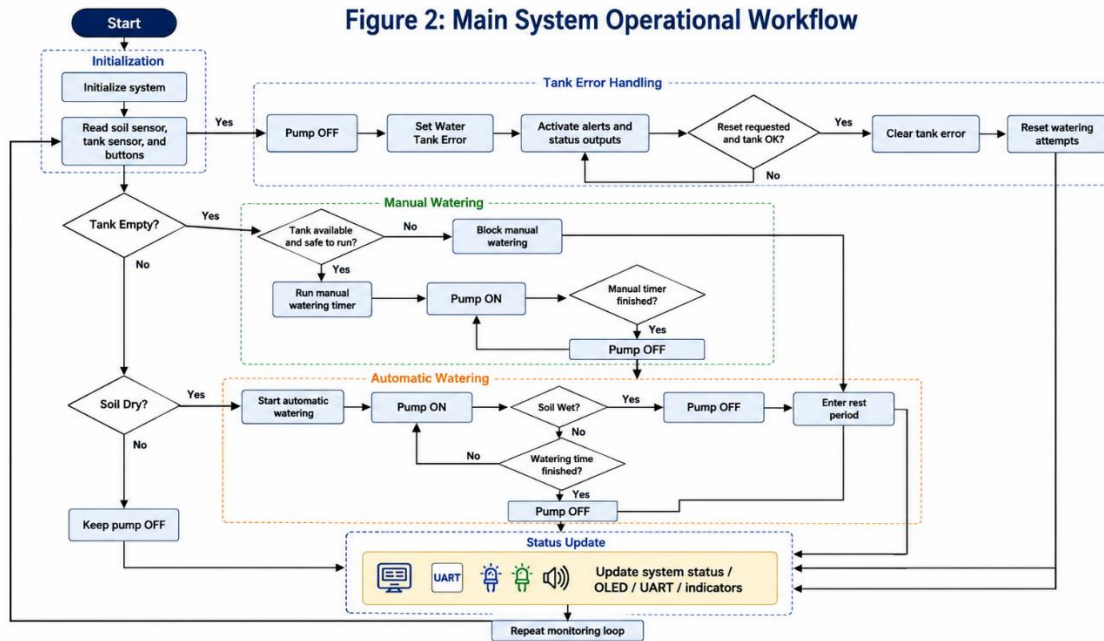
This document defines what the software shall do. Detailed internal design decisions, source-code structure, and algorithm implementation details are documented separately in the System Design Document.

3. Software Scope

The software shall implement one complete irrigation control unit with the following capabilities:

- Read soil moisture using ADC.
- Convert ADC reading into moisture percentage.
- Classify soil as WET or DRY.
- Monitor water tank availability.
- Control one pump or relay output.
- Support automatic watering.

- Support timed manual watering.
- Protect the pump when water is unavailable.
- Limit repeated watering attempts.
- Detect invalid ADC readings.
- Report the active system error.
- Update OLED display information.
- Send UART status reports.
- Control heartbeat LED, error LED, and buzzer.
- Use CMSIS-RTOS V2 tasks, queues, event flags, and mutexes.



4. System Context

The Smart Irrigation software acts as the control layer between physical inputs and system outputs.

The software receives data from:

- Soil moisture sensor through ADC.
- Water tank level sensor through GPIO.
- Manual watering button through GPIO.
- Reset button through GPIO.

The software controls:

- Pump or relay output.
- Heartbeat LED.
- Error LED.
- Buzzer.
- SSD1306 OLED display through I2C.
- UART terminal through USART.

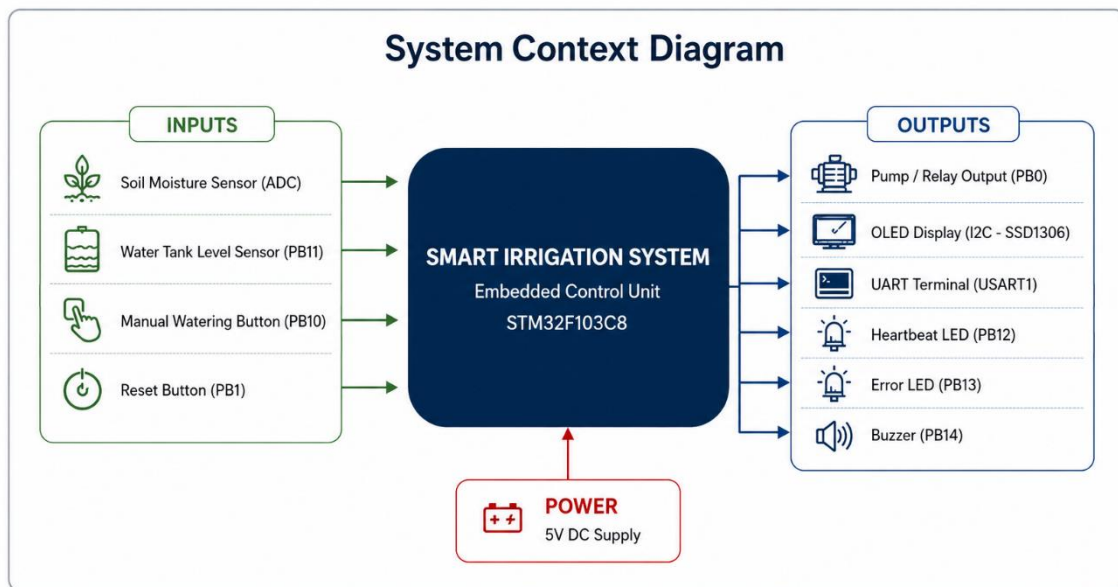


Figure 1: System Context Diagram

5. Software Interfaces

5.1 ADC Interface Requirements

ID	Requirement
SIF-ADC-01	The software shall read soil moisture through ADC1.
SIF-ADC-02	The software shall obtain an ADC value within the range 0 to 4095.
SIF-ADC-03	The software shall detect ADC readings outside the configured valid range.
SIF-ADC-04	The software shall store the latest valid ADC value for reporting.

5.2 GPIO Input Interface Requirements

ID	Requirement
SIF-GPIO-01	The software shall read the reset button through PB1.
SIF-GPIO-02	The software shall read the manual watering button through PB10.
SIF-GPIO-03	The software shall read the water tank sensor through PB11.
SIF-GPIO-04	The software shall treat a pressed button as GPIO_PIN_RESET.
SIF-GPIO-05	The software shall treat GPIO_PIN_RESET on PB11 as water available.
SIF-GPIO-06	The software shall treat GPIO_PIN_SET on PB11 as water low or tank empty.

5.3 GPIO Output Interface Requirements

ID	Requirement
SIF-OUT-01	The software shall control the pump or relay through PB0.
SIF-OUT-02	The software shall control the heartbeat LED through PB12.
SIF-OUT-03	The software shall control the error LED through PB13.
SIF-OUT-04	The software shall control the buzzer through PB14.
SIF-OUT-05	The pump output shall be controlled only by the pump control software module.

5.4 UART Interface Requirements

ID	Requirement
SIF-UART-01	The software shall send periodic system reports through USART1.
SIF-UART-02	UART reports shall include current state, current error, ADC value, moisture percentage, soil status, pump status, tank status, and watering attempts.
SIF-UART-03	UART transmission shall be protected from concurrent access.
SIF-UART-04	The software may provide optional raw ADC debug output.

5.5 I2C and OLED Interface Requirements

ID	Requirement
SIF-I2C-01	The software shall communicate with the SSD1306 OLED display through I2C1.
SIF-I2C-02	OLED communication shall use PB6 as SCL and PB7 as SDA.
SIF-I2C-03	The OLED shall display system state, moisture percentage, soil status, pump status, tank status, attempts, and current error.
SIF-I2C-04	I2C access shall be protected from concurrent access.
SIF-I2C-05	The software shall report OLED/I2C communication faults when detected.

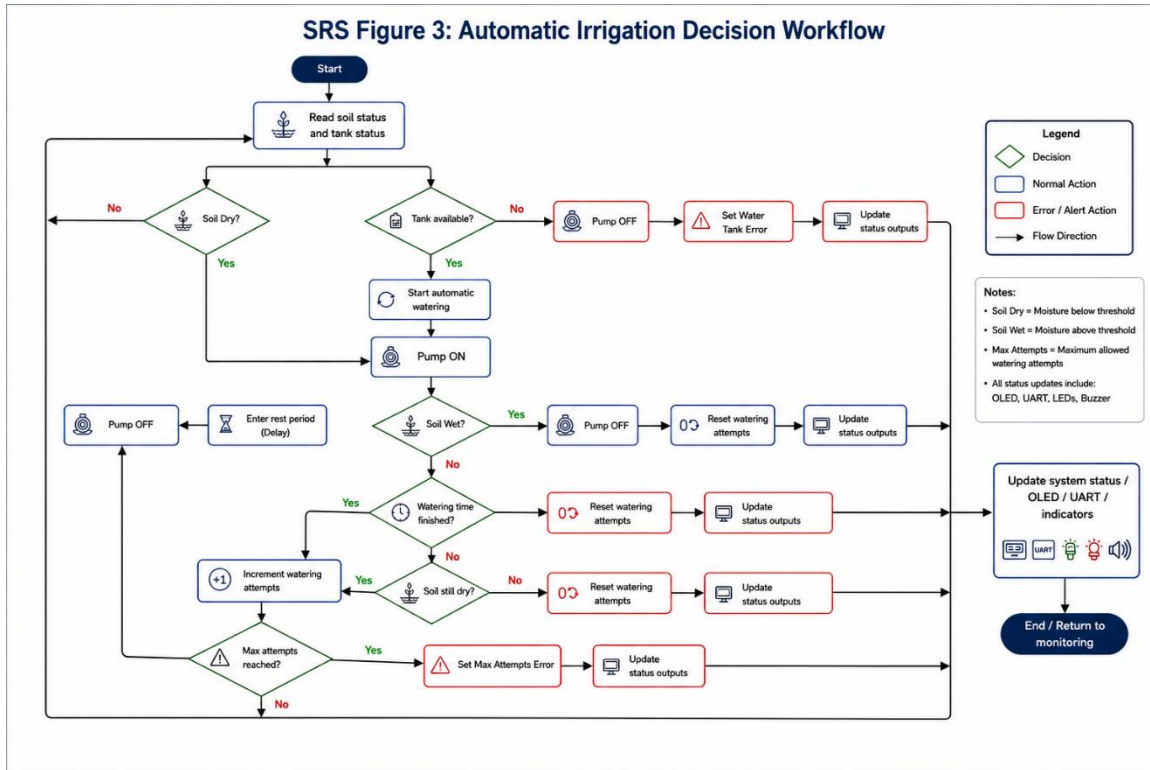
6. Software Functional Requirements

6.1 Soil Measurement Requirements

ID	Requirement
SFR-01	The software shall read soil ADC data periodically.
SFR-02	The software shall calculate moisture percentage from the ADC reading.
SFR-03	The software shall classify soil as DRY when ADC value is greater than SOIL_DRY_THRESHOLD.
SFR-04	The software shall classify soil as WET when ADC value is lower than SOIL_WET_THRESHOLD.
SFR-05	The software shall preserve the previous soil condition when ADC value is between the wet and dry thresholds.
SFR-06	The software shall generate a dry-soil notification only when soil state changes to DRY.
SFR-07	The software shall generate a wet-soil notification only when soil state changes to WET.

6.2 Automatic Irrigation Requirements

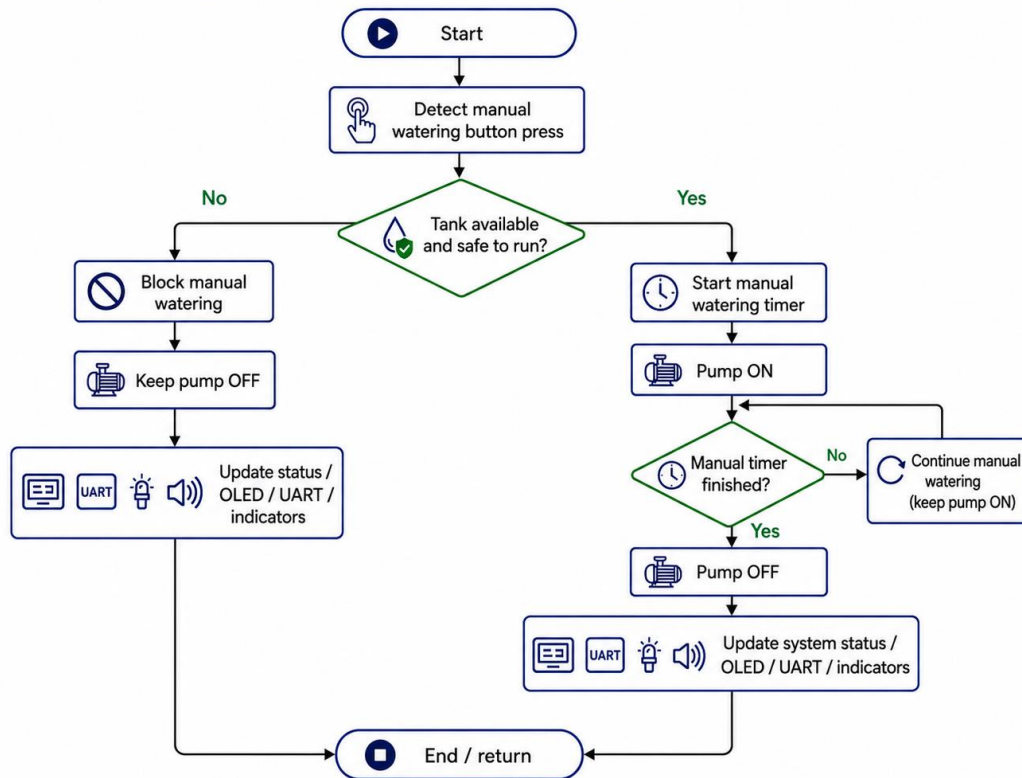
ID	Requirement
SFR-08	The software shall begin automatic watering when soil becomes DRY and water tank status is OK.
SFR-09	The software shall turn the pump ON during automatic watering.
SFR-10	The software shall stop the pump when soil becomes WET during automatic watering.
SFR-11	The software shall stop the pump when the configured automatic watering duration expires.
SFR-12	The software shall enter a rest period after each automatic watering cycle.
SFR-13	The software shall check soil condition after the rest period.
SFR-14	The software shall repeat watering only when soil remains DRY and attempts are below the maximum limit.
SFR-15	The software shall increment the watering attempts counter after a failed watering cycle.
SFR-16	The software shall reset watering attempts when soil becomes WET.
SFR-17	The software shall enter maximum attempts error state when soil remains DRY after the maximum allowed attempts.



6.3 Manual Watering Requirements

ID	Requirement
SFR-18	The software shall detect a valid manual watering button press.
SFR-19	The software shall reject repeated false manual button activations caused by button bounce.
SFR-20	The software shall start manual watering only when tank status is OK.
SFR-21	The software shall turn the pump ON during manual watering.
SFR-22	The software shall keep the pump ON for <code>MANUAL_WATER_TIME_MS</code> .
SFR-23	The software shall turn the pump OFF when manual watering time expires.
SFR-24	The software shall block manual watering when tank status is EMPTY.

Figure 4: Manual Watering Workflow



6.4 Water Tank Protection Requirements

ID	Requirement
SFR-25	The software shall monitor tank status periodically.
SFR-26	The software shall generate a water-low notification when the tank becomes EMPTY.
SFR-27	The software shall immediately turn the pump OFF when water-low condition is detected.
SFR-28	The software shall enter WATER_TANK_EMPTY state when the tank is empty.
SFR-29	The software shall set SYSTEM_ERROR_WATER_TANK when the tank is empty.
SFR-30	The software shall keep water tank error active after tank recovery until reset is requested.
SFR-31	The software shall clear water tank error only when tank status is OK and reset button is pressed.

6.5 Error Handling Requirements

ID	Requirement
SFR-32	The software shall detect invalid ADC readings.
SFR-33	The software shall set SYSTEM_ERROR_ADC_SENSOR when ADC reading is outside the valid configured range.
SFR-34	The software shall detect maximum watering attempts failure.
SFR-35	The software shall set SYSTEM_ERROR_MAX_ATTEMPTS when the maximum attempts condition occurs.
SFR-36	The software shall support OLED/I2C communication error reporting.
SFR-37	The software shall maintain independent active flags for supported error types.
SFR-38	The software shall determine the highest-priority active error.
SFR-39	The software shall prioritize SYSTEM_ERROR_WATER_TANK above all other errors.
SFR-40	The software shall prioritize SYSTEM_ERROR_MAX_ATTEMPTS above ADC and OLED/I2C errors.
SFR-41	The software shall update the current system error after any error set or clear operation.
SFR-42	The software shall send system error or system OK indication based on the current active error state.

6.6 Monitoring and Reporting Requirements

ID	Requirement
SFR-43	The software shall toggle the heartbeat LED periodically.
SFR-44	The software shall turn the error LED ON when a system error is active.
SFR-45	The software shall turn the error LED OFF when no system error is active.
SFR-46	The software shall activate the buzzer during active system errors.
SFR-47	The software shall disable the buzzer when no error is active.
SFR-48	The software shall create a complete status report periodically.
SFR-49	The software shall send status reports to UART reporting and OLED display independently.
SFR-50	The software shall include active error information in all status outputs.

7. RTOS Requirements

ID	Requirement
SRTOS-01	The software shall use CMSIS-RTOS V2 API.
SRTOS-02	The software shall use separate tasks for soil sensing, pump control, tank monitoring, button handling, system monitoring, UART reporting, and OLED updates.
SRTOS-03	The software shall assign pump control higher priority than display and reporting tasks.
SRTOS-04	The software shall use event flags for soil, tank, manual, reset, and system state notifications.
SRTOS-05	The software shall use queues for structured status data transfer.
SRTOS-06	The software shall use separate status queues for UART reporting and OLED updates.
SRTOS-07	The software shall use mutexes to protect UART and I2C resources.
SRTOS-08	The software shall avoid direct pump control from sensor, button, UART, or OLED tasks.
SRTOS-09	The software shall centralize irrigation decisions in PumpControlTask.

8. Data Requirements

8.1 Soil Data Requirements

Data Item	Description
adcValue	Latest ADC reading from soil sensor.
moisturePercent	Calculated soil moisture percentage.
soilStatus	WET or DRY classification.

8.2 System Status Requirements

The software shall maintain a complete system status record containing:

Data Item	Description
adcValue	Latest soil ADC value.
moisturePercent	Current calculated moisture percentage.
soilStatus	Current soil condition.
pumpStatus	Current pump ON/OFF state.

Data Item	Description
waterTankStatus	Tank OK or EMPTY state.
wateringAttempts	Current automatic watering attempt count.
currentState	Current irrigation state.
currentError	Highest-priority active system error.

9. Timing Requirements

ID	Requirement
STIME-01	Soil sensor readings shall be performed periodically.
STIME-02	Water tank state shall be monitored periodically.
STIME-03	Button state shall be monitored periodically with debounce support.
STIME-04	Automatic watering duration shall be limited by PUMP_WATERING_TIME_MS.
STIME-05	Rest duration shall be limited by PUMP_REST_TIME_MS.
STIME-06	Manual watering duration shall be limited by MANUAL_WATER_TIME_MS.
STIME-07	System reports shall be generated periodically according to SYSTEM_STATUS_REPORT_PERIOD_MS.
STIME-08	Heartbeat LED shall toggle periodically.
STIME-09	Buzzer shall toggle periodically while an error is active.

10. Error Handling and Safety Requirements

ID	Requirement
SAFE-01	The software shall force pump OFF when water tank is empty.
SAFE-02	The software shall force pump OFF when maximum attempts error is active.
SAFE-03	The software shall stop pump when soil becomes wet during watering.
SAFE-04	The software shall prevent manual watering when tank is empty.
SAFE-05	The software shall maintain water tank error until tank recovery and user reset.
SAFE-06	The software shall report critical errors through OLED, UART, LED, and buzzer.
SAFE-07	The software shall give water tank error the highest priority.

ID	Requirement
SAFE-08	The software shall centralize all pump decisions inside PumpControlTask.
SAFE-09	The software shall preserve safe pump OFF behavior during error conditions.

11. User Interface Requirements

ID	Requirement
SUI-01	The OLED shall display current irrigation state.
SUI-02	The OLED shall display soil moisture percentage.
SUI-03	The OLED shall display pump status.
SUI-04	The OLED shall display tank status.
SUI-05	The OLED shall display watering attempts.
SUI-06	The OLED shall display current error.
SUI-07	UART shall provide readable periodic system reports.
SUI-08	The heartbeat LED shall indicate that the software is running.
SUI-09	The error LED shall indicate active error condition.
SUI-10	The buzzer shall provide audible error indication.

12. Performance and Reliability Requirements

ID	Requirement
SPR-01	Pump safety responses shall have higher execution priority than reporting functions.
SPR-02	The system shall avoid unstable WET/DRY switching by using hysteresis behavior.
SPR-03	The system shall prevent repeated button actions caused by bouncing.
SPR-04	UART output shall not become corrupted due to concurrent task access.
SPR-05	I2C OLED communication shall not become corrupted due to concurrent task access.
SPR-06	The software shall continue heartbeat operation while normal scheduling is active.
SPR-07	The system shall maintain valid status reporting during normal and error conditions.

13. Technical Constraints

ID	Constraint
TC-01	The software shall run on STM32F103C8.
TC-02	The software shall use STM32 HAL drivers.
TC-03	The software shall use CMSIS-RTOS V2.
TC-04	The soil sensor shall be connected through ADC1.
TC-05	The OLED shall use SSD1306 over I2C1.
TC-06	UART reporting shall use USART1.
TC-07	The water tank sensor shall use PB11.
TC-08	Manual watering button shall use PB10.
TC-09	Reset button shall use PB1.
TC-10	Pump control output shall use PB0.
TC-11	The simulation shall use Proteus.
TC-12	The pump shall be represented by relay output or LED in simulation.

14. Software Requirements Traceability Matrix

CRS Requirement Area	Software Requirement Group	Verification / Validation Evidence
Soil monitoring	SFR-01 to SFR-07	UT-01, UT-03, IT-01, BV-01 to BV-03
Automatic watering	SFR-08 to SFR-17	ST-02, ST-03, ST-04, BV-02 to BV-05
Manual watering	SFR-18 to SFR-24	IT-03, ST-08, ST-09, BV-08, BV-09
Tank protection	SFR-25 to SFR-31	IT-02, ST-06, ST-07, BV-06, BV-07
Error handling	SFR-32 to SFR-42	UT-02, UT-04, ST-10, ST-11, BV-10, BV-11
System monitoring	SFR-43 to SFR-50	IT-04, IT-05, IT-06, BV scenarios
RTOS behavior	SRTOS-01 to SRTOS-09	Integration and system tests
Safety requirements	SAFE-01 to SAFE-09	ST-03 to ST-11, BV-03 to BV-11
User interface	SUI-01 to SUI-10	IT-04, IT-05, all Proteus validation tests

15. Conclusion

This Software Requirements Specification defines the detailed software behavior required for the Smart Irrigation System. It provides a clear implementation baseline for sensor processing, irrigation control, safety behavior, RTOS usage, data reporting, user interaction, and error management.

The requirements in this document will be used as the baseline for the System Design Document, firmware implementation, verification, and black-box validation.